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NEUROPIXEL DATA TO 3D VISUALISATION

BME 4509 Group project
Team:Imperial PixelMinds

scan for more Info from our website including our report

With heartfelt thanks to Professor Claudia Clopath and Dr. Albert Albesa Gonzalez for their support.

IMPERIAL

MOTIVATION

The brain has been a blackbox of unknowns for centuries. At its core, we still don't know exactly **how networks of neurons work together** to turn what we see and think into behavior. Understanding **how neurons function and communicate** can significantly improve our understanding of various psychological and physiological conditions affecting the human brain.

- ⊕ High-resolution of neuron architecture
- ⊖ Time consuming and labour intensive

Today's Neuropixel probes give us lots of detailed 2D action potential and position data, but turning that into a true 3D map of brain activity means slow, manual steps—cutting and staining tissue after the experiment and hand-sorting measured action potentials.

AIM

Building on the framework of Steinmetz et al. and modern spike-localization methods we aimed to;

- Automatically label neurons based on what stimuli they respond to.
- Reconstruct 3D spike locations from the probe's 2D recordings
- Map those neurons onto a standard 3D mouse-brain atlas

FUTURE POTENTIAL

Spiking data analysis pipeline and results

Input: Spike events
→ Step 1: Assign cluster ID
For each spike:
Assign to a cluster → Step 2: Statistical selection
For each cluster:
Compute stimulus-response significance
If p-value < threshold:
Mark as significantly responsive
→ Step 3: Filtering
Filter out:
- Non-significant clusters
- Low firing rate neurons
- Noise/artifacts
Output: Set of significant, high-quality neuronal clusters

Session 1

Output 1: PSTH plotting

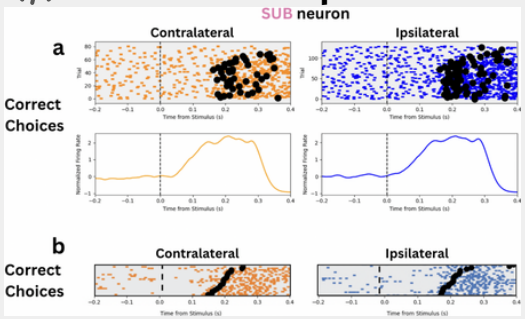
Session 2

→ Step 4: Constructing the Toeplitz matrix
Build matrix for the three-neuron function
→ Step 5: ElasticNet regression
Use ElasticNet to find triple kernel for neuron response modeling
→ Step 6: Nested test
Identify the kernel type (visual / motor / mixed)
Output 2: Plot of predicted neuron activity according to kernels

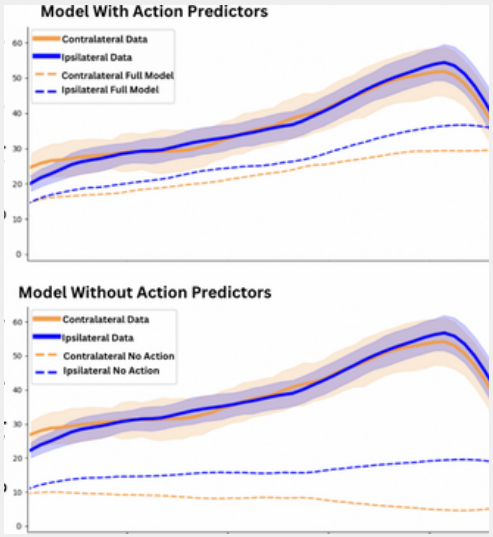
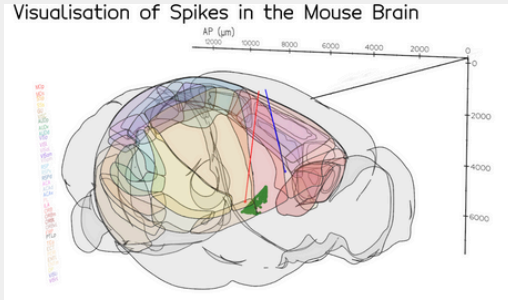
Session 3

→ Step 4: PTP calculation
Input 2: Channel coordinates
→ Step 5: Spike localisation
→ Step 6: Smoothing by AMSGrad optimiser
Output 3: Visualisation of 3D spikes in brain render

Session 1 Output



Session 2 Output



Session 3 Output